Optimizing Diabetes Care

Balancing Intervention Costs and Readmission Reduction

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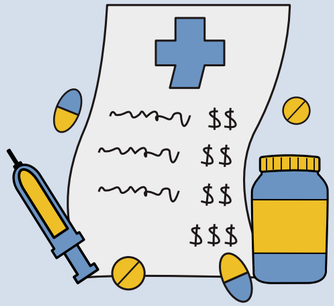
Agenda

1. PROBLEM STATEMENT & MOTIVATION
2. HOSPITAL CARE WORKFLOW:
3. DATA OVERVIEW AND PREPROCESSING
4. ML PIPELINE
   - MODEL PREDICTION
   - MODEL EVALUATION
5. OPTIMIZATION
   - MODEL
   - RESULTS AND ANALYSIS
6. LIMITATIONS AND ASSUMPTIONS
7. TRADE-OFFS, ALTERNATIVES, AND KEY INSIGHTS
8. FUTURE SCOPE

Problem Statement

Hospital readmissions remain a major driver of rising healthcare costs, adding financial strain to hospitals and the broader healthcare system. Among these, diabetes-related readmissions represent a particularly persistent challenge because of their prevalence and cost impact.

Problem Context and Motivation



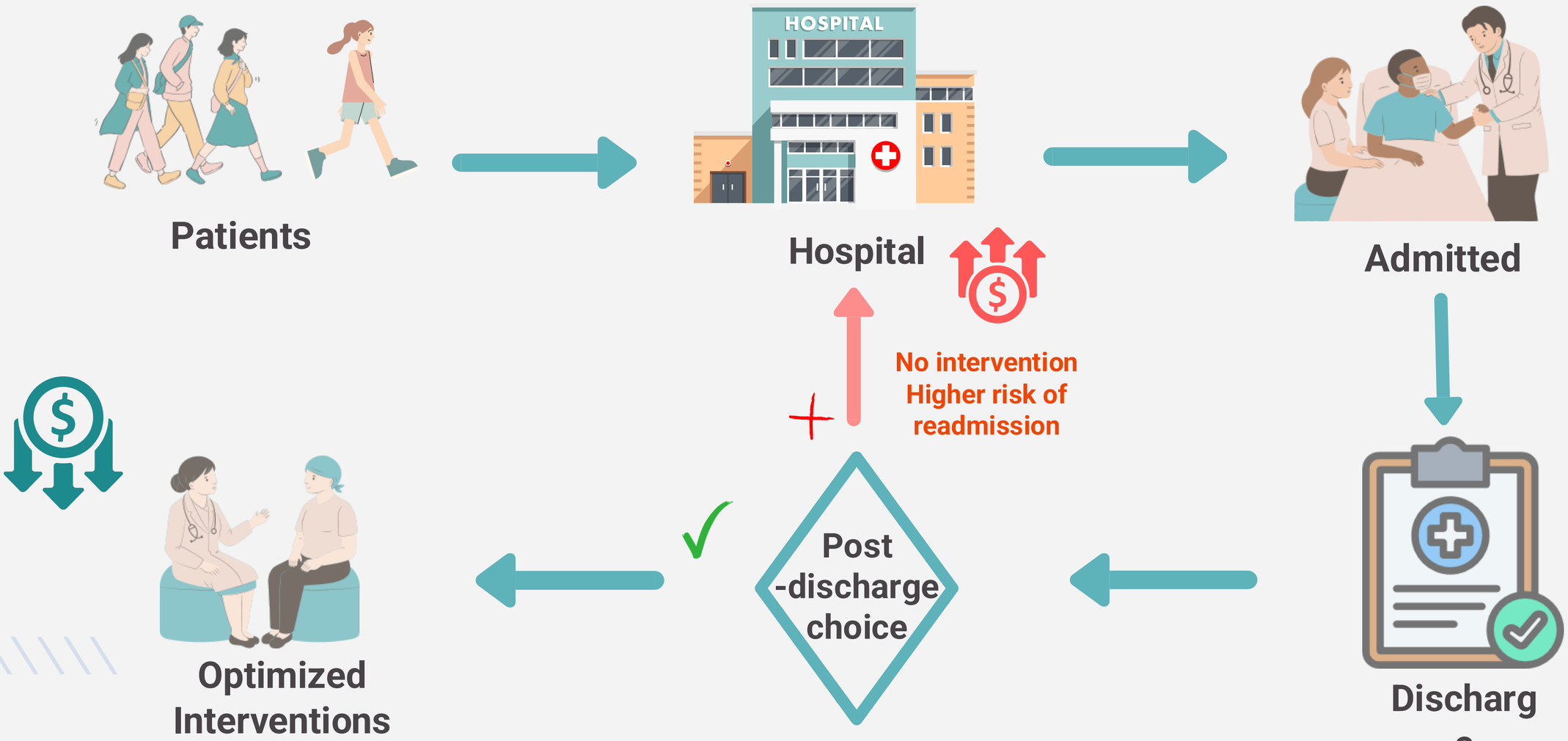
Hospital readmissions drive up healthcare spending accounting for $52.4 billion and contributing to the $4.1 trillion annual cost of U.S. healthcare.

Hospital readmissions among patients with diabetes are a significant and growing burden on healthcare systems.

Diabetes accounted for about 32% of all readmissions and generated approximately $12.8 billion in hospital costs in 2018.

Hospital Care Workflow:

Default vs. Predict-Then-Optimize Approach



Data Overview & Preprocessing

Encounter-Level Diabetic Patients' Re-admission from UC Irvine ML Repository

- 100,000+ encounters with 50 columns
- Key columns: patient demographics, medications, number of procedures, diagnoses

Mapped coded variables to readable labels

```
admission_type_id,description
1,Emergency
2,Urgent
3,Elective
4,Newborn
5,Not Available
6,NULL
7,Trauma Center
8,Not Mapped
,
discharge_disposition_id,description
1,Discharged to home
2,Discharged/transferred to another short term hospital
3,Discharged/transferred to SNF
4,Discharged/transferred to ICF
```

Handled missing values

weight	96.858479
max_glu_serum	94.746772
A1Cresult	83.277322
medical_specialty	49.093017
payer_code	39.557416
race	2.233555
diag_3	1.398306
diag_2	0.351787
diag_1	0.020636

Removed columns with ~ 40% missing values

ML Pipeline

Removed patient_nbr

3. Correlation Analysis Results:

	feature	correlation	p_value
8	number_inpatient	0.214869	0.000000e+00
9	number_diagnoses	0.112523	3.804819e-228
7	number_emergency	0.099609	5.432651e-179
6	number_outpatient	0.081954	1.271938e-121
1	patient_nbr	0.072012	2.806772e-94
11	diabetesMed	0.062834	9.771385e-72
2	time_in_hospital	0.051360	8.396942e-49

5.2 Categorical Feature Analysis

Chi-Square Test with Cramer's V Results:

	feature	chi2	p_va
3	diag_1	2929.429477	1.997371e-
4	diag_2	2479.133253	9.597057e-
5	diag_3	2406.884371	1.234104e-
28	discharge_disposition_desc	1921.413581	0.000000e-
29	admission_source_desc	858.266263	3.751192e-
21	insulin	407.674443	4.817012e-
27	admission_type_desc	234.590022	1.109811e-

Train/Test Split

- Perform Group-Based Split:
  - Why: Prevents data leakage by ensuring no patient appears in both training and test sets
  - How: Uses GroupShuffleSplit with patient_nbr as the grouping variable
  - Method: gss = GroupShuffleSplit(n_splits=1, test_size=0.2, random_state=42)

Numerical Feature Analysis

Categorical Feature Analysis

Model Preparation (feature selection)

- Select features for both numerical and catergorical ones

```
Numerical Features (6):
- number_inpatient
- number_diagnoses
- number_emergency
- number_outpatient
- diabetesMed
- time_in_hospital

Categorical Features (7):
- diag_1
- diag_2
- diag_3
- discharge_disposition_desc
- admission_source_desc
- insulin
- admission_type_desc
```

Model Training

Four Models:

1. Logistic Regression - Linear baseline
2. Random Forest - Ensemble method
3. XGBoost - Gradient boosting
4. HistGradientBoosting - Scikit-learn boosting

Model Evaluation

Cross-Validation

ROC, AUC

Prediction Model

Four Models

- **Logistic Regression** - Linear baseline
- **Random Forest** - Ensemble method
- **XGBoost** - Gradient boosting
- **HistGradientBoosting** - Scikit-learn boosting

Cross-Validation Setup

- Method: 5-fold StratifiedGroupKFold
- Preprocessing: StandardScaler + OneHotEncoder

Cross-Validation Results:

	Log Loss	Brier Score	ROC AUC	Average Precision
Logistic Regression	0.632 ± 0.002	0.222 ± 0.001	0.686 ± 0.001	0.646 ± 0.006
Random Forest	0.644 ± 0.003	0.225 ± 0.001	0.681 ± 0.003	0.629 ± 0.003
XGBoost	0.625 ± 0.002	0.219 ± 0.001	0.695 ± 0.002	0.654 ± 0.003
HistGradientBoosting	0.624 ± 0.002	0.219 ± 0.001	0.697 ± 0.002	0.657 ± 0.005

Test Set Results:

	Log Loss	Brier Score	ROC AUC	Average Precision
Logistic Regression	0.627449	0.220085	0.688332	0.644824
Random Forest	0.647073	0.224934	0.679073	0.617306
XGBoost	0.622565	0.217877	0.69651	0.653833
HistGradientBoosting	0.621731	0.217481	0.698549	0.654131

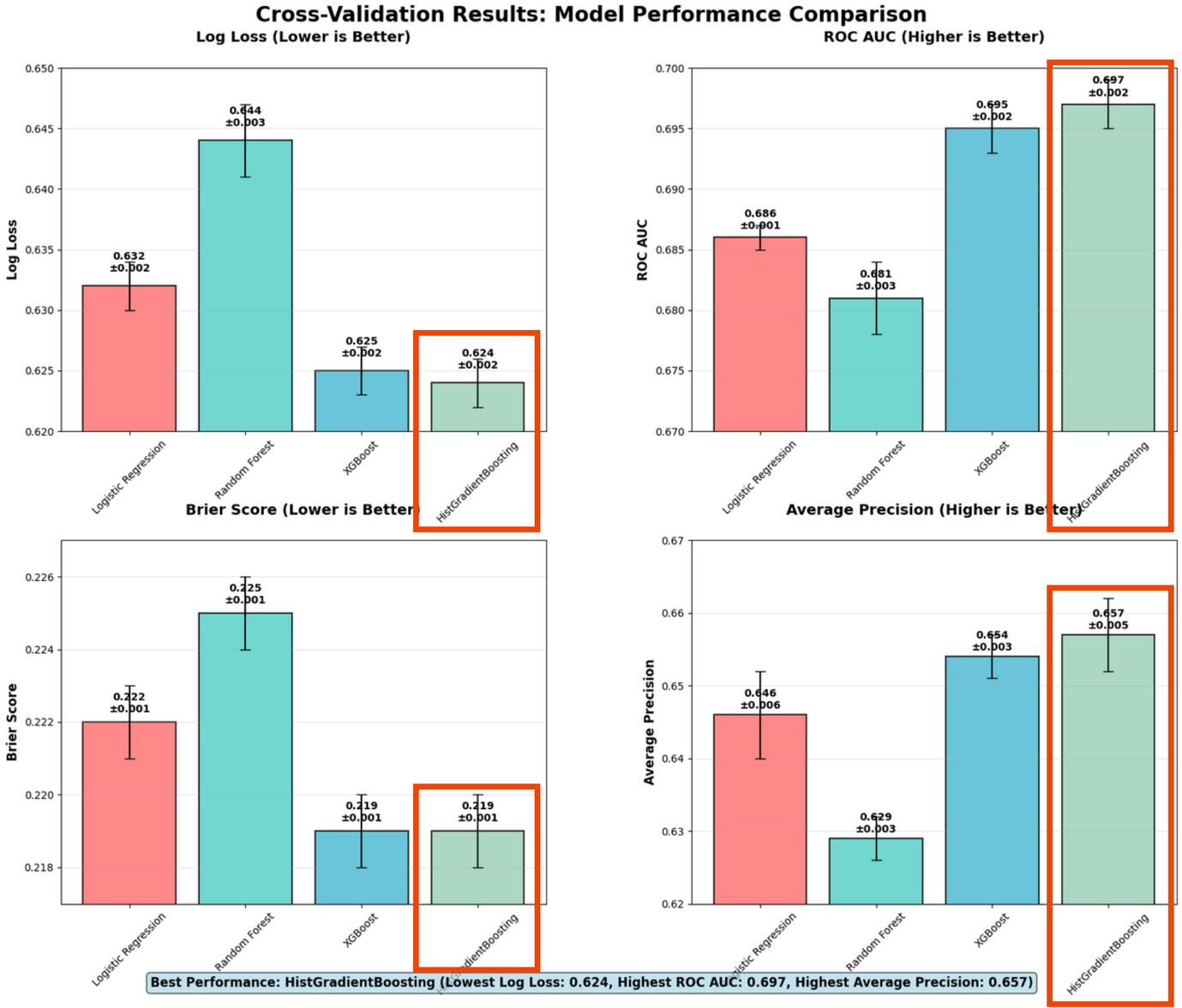
Metrics

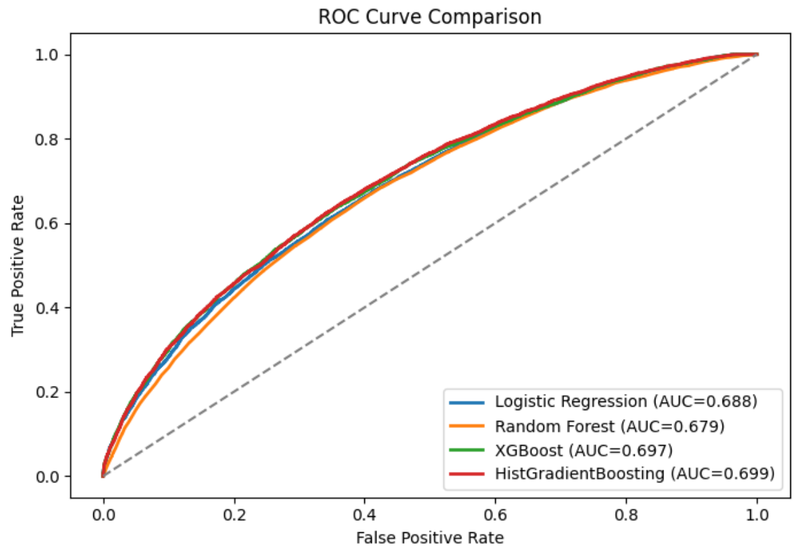
- **Log Loss** (0.624 - 0.644)
  Lower values = better calibrated probabilities
- **ROC & AUC** (0.681 - 0.697)
  0.7+ = Good discrimination ability to distinguish readmission vs. no readmission
- **Brier Score** (0.219 - 0.225)
  Lower values = better probability calibration
- **Average Precision (0.629 - 0.657)**
  Higher values = better at identifying high-risk patients

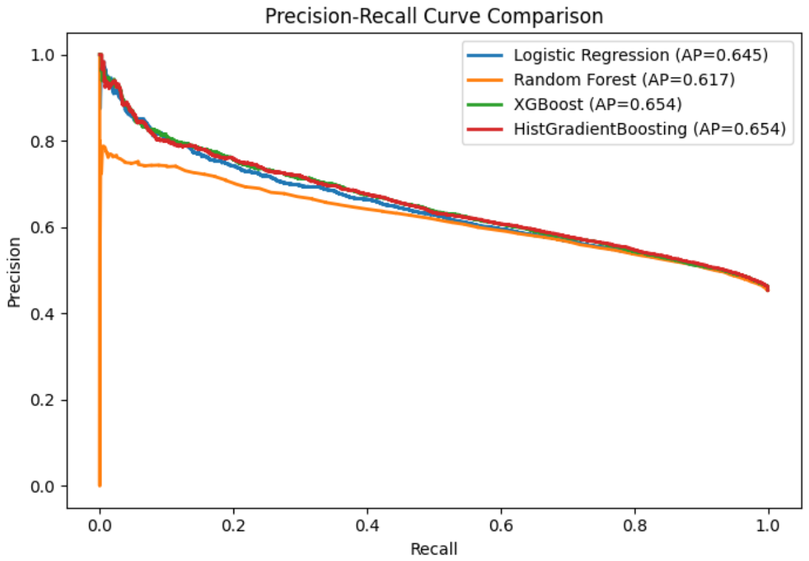
Summary

- **Model Performance**: HistGradientBoosting achieves best overall performance
- **Strong Discrimination**: All models achieve ROC AUC > 0.68
- **Good Calibration**: Log loss < 0.65 across all models
- **Consistent Performance**: Cross-validation and test results align well

Model Evaluation







Optimization Model: Knapsack

Decision Variables

$$Xi$$

1 if the hospital performs an intervention for encounter i;
0 otherwise

Optimization Model: Knapsack

Objective Functions:

Maximize cost savings through reduced readmissions

- **Probability of readmission**
- **Readmission costs (by age groups)**
- **Intervention cost (by age groups)**
  e.g. follow-up care, patient education, or post-discharge monitoring
- **Intervention effectiveness** (reduction in readmission probability)

$$\max \sum_{i=1}^{n} X_i \left(e * P_{a(i)} * C_{a(i)} - C_{int_{a(i)}} \right)$$

Optimization Model: Knapsack

Constraints

Budget limit

$$\sum_{i=1}^{n} X_i * Cost_{(i)} \leq B$$

Binary constraint

$$X_i \in \{0,1\},\ i=1,\ldots,n.$$

Optimized Result

encounter_id	...	readmission_cost	selected_for_intervention
42570	...	13000	1.0
182796	...	11000	-0.0
253380	...	9500	-0.0
260166	...	11000	-0.0
377268	...	8000	-0.0
...	...	...	...
443816024	...	11000	-0.0
443824292	...	13000	-0.0
443841992	...	7000	-0.0
443842070	...	9500	-0.0
443842778	...	13000	-0.0

Under $1M budget...

- **1832** patients are selected for interventions
- With the total objective value (reduced projected readmission costs) being **2.2 M**

To allocate limited intervention resources toward patients with the greatest potential cost savings!

Limitations and Assumptions

- **Sparse clinical fields:** key labs/specialty (e.g., A1Cresult, max_glu_serum, weight) are heavily missing → removed; this can bias signals toward administrative variables.
- **Cost inputs:** readmission cost Cr and intervention cost Cint are assumed to be fixed within each age group, based on trends reported in published research.
- **Effectiveness:** a single (or age-group) intervention effectiveness reduces risk proportionally for all treated patients.
- **Equity/coverage:** no explicit fairness or service-level constraints, allocation is purely driven by expected savings.

Trade-Offs, Alternatives, and Key Insights

Trade-offs

- **Immediate vs. Long-Term Impact:** Targeting older age groups yields larger short-term cost savings, while investing in younger patients may deliver long-term system-wide benefits.

- **Precision vs. Equity:** Prioritizing predicted high-risk patients maximizes cost reduction but could overlook moderate-risk groups who also benefit from intervention.

Alternatives

- **Multi-Armed Bandits (MAB):** Adaptive Intervention Allocation
  - Instead of fixed optimization, a contextual bandit could learn dynamically which intervention yields the best cost-savings per patient type over time, balancing exploration and exploitation.

- **Reinforcement Learning (RL) Optimization:**
  - Frame intervention allocation as a sequential decision process where the agent learns a policy that maximizes cumulative cost savings under changing patient states and budgets.

Trade-Offs, Alternatives, and Key Insights

Key Insights for Decision-Makers

- **Optimization shifts outcomes:** A predict-then-optimize framework transforms decision-making from reactive (treat after readmission) to proactive (prevent it), directly linking analytics to financial savings.

- **Not all readmissions are equal:** Intervening on every patient is not cost-effective, targeted allocation based on predictions maximizes impact.

- **Optimization Creates Policy Levers:** You can tune constraints (budget, equity floors, capacity) to reflect hospital priorities and see transparent trade-offs in dollars saved.

Future Scope

- **Incorporate richer clinical features:** Integrate lab values, vitals, and longitudinal patient history to improve prediction accuracy.
- **Model multi-tier interventions:** Move beyond binary decisions to allow multiple intervention intensities and types.
- **Add operational constraints:** Include staff capacity, weekly scheduling, and service-level requirements in the optimization.
- **Equity-focused optimization:** Add fairness constraints to ensure balanced intervention allocation across demographic groups.

THANK YOU